



AI SOFTWARE DEVELOPMENT INTERNSHIP

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Week 5 — Detailed Task Sheet

Embeddings & Vector Databases • RAG Fundamentals Manual Retrieval-
Augmented Generation Pipeline • Git — `git revert`

PART B — VECTOR DATABASES

Practice Tasks

1. Run the script above and confirm the space-related query returns the science fiction book first.

Outputs:

```
PS D:\Week5\Week5_PartB_Chroma> python chroma_demo.py
Query: a space journey story
Documents:
[['A crew travels through a wormhole to save humanity from a dying Earth.', 'A young wizard attends a magic school and fights a dark lord.']]
Metadata:
[[{'source': 'book_2.txt', 'category': 'Science Fiction'}, {'category': 'Fantasy', 'source': 'book_1.txt'}]]
```

2. Add a metadata filter to the query (where={"category": "Fantasy"}) and confirm it only searches within that category.

Output:

```
PS D:\Week5\Week5_PartB_Chroma> python chroma_demo.py
Query: a space journey story
Documents:
[['A young wizard attends a magic school and fights a dark lord.']]
Metadata:
[[{'category': 'Fantasy', 'source': 'book_1.txt'}]]
```

3. Add two more documents of your own and re-run a few queries to get comfortable with the add/query pattern.

Output:

```
Problems Output Debug Console Terminal Ports
PS D:\Week5\Week5_PartB_Chroma> python chroma_demo.py
Query: a space journey story
Documents:
[['A group of astronauts explores a distant planet.', 'A crew travels through a wormhole to save humanity from a dying Earth.']]
Metadata:
[[{'source': 'book_5.txt', 'category': 'Science Fiction'}, {'source': 'book_2.txt', 'category': 'Science Fiction'}]]
PS D:\Week5\Week5_PartB_Chroma> ^C
```